

Piecewise, Absolute-Value and Linear-Quadratic Models — Boundaries and Endpoints

Answer Key

Algebra 1

Quick Answers

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|------|---------------------------------|
| 1. 3 | 5. 0 |
| 2. 5 | 6. 1, 4 |
| 3. 7 | 7. -5 |
| 4. 5 | 8. $x = 1, y = 0, x = 4, y = 3$ |
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Curriculum

Level: Algebra 1HSA-CED.A.1–A.4 – *problems 6, 8*HSF-IF.A–HSF-IF.C – *problems 1, 2, 3, 4, 5, 7**Difficulty 1–5 of 5 across 9 tagged checks.*

Common wrong answers

2. *If they got 7: used the top rule at the boundary, but that rule is written with a strict inequality and so does not own the boundary point*
3. *If they got 5: took the boundary to belong to the second rule, but this time the first rule carries the closed inequality*
4. *If they got 3: read the boundary off the second rule, which the overlapping definition also allows - two outputs for one input is exactly the bug*
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1. Which rule owns $x = 1$ for F , and what is $F(1)$?

Step 1: test the input against each condition, not against the formulas. Is $1 < 3$? Yes, so the top rule owns $x = 1$. Is $1 \geq 3$? No, so the bottom rule is out.

Step 2: substitute into the rule that owns the input: $2(1) + 1$.

Step 3: evaluate:

3

Note the order of operations that matters most here: choose the rule first, substitute second. Choosing by “which formula looks easier” is what breaks in problem 2.

2. Mia writes $F(3) = 7$. Which rule did she use, why is it wrong, and what is $F(3)$?

Step 1: identify Mia's rule. $2(3) + 1 = 7$, so she used the top rule. The top rule carries $x < 3$, a *strict* inequality, which excludes the boundary itself — it holds for 2.9 and 2.999 but never for 3.

Step 2: find the rule that does own $x = 3$. The bottom rule carries $x \geq 3$, a closed inequality, and $3 \geq 3$ is true.

Step 3: substitute into the bottom rule: $3^2 - 4 = 9 - 4$:

5

Step 4: name the misconception for the student: the top rule's values *approach* 7 as x climbs toward 3, but the value *at* 3 is 5. A limit approached and a value taken are different things, and the inequality sign is what tells them apart.

3. Leo writes $G(-2) = 5$ using “the second rule at the boundary”. What is wrong, and what is $G(-2)$?

Step 1: name the faulty habit. Leo has generalised from a function like F where the boundary happened to sit in the second rule. Which rule owns a boundary is not a fixed convention — it is whichever rule carries the closed inequality, and that can be either one.

Step 2: read this function's conditions. The top rule carries $x \leq -2$, which is closed, and $-2 \leq -2$ is true. The bottom rule carries $x > -2$, strict, and $-2 > -2$ is false. The top rule owns the boundary here.

Step 3: substitute into the top rule: $5 - (-2) = 5 + 2$:

7

Step 4: check what Leo's value was. $(-2)^2 + 1 = 5$ is the bottom rule's output, which is the value G approaches from the right but never takes.

4. H assigns two outputs to $x = 1$. Repair it so the top rule keeps the boundary, and give $H(1)$.

Step 1: diagnose. At $x = 1$ the top condition $x \leq 1$ is true *and* the bottom condition $x \geq 1$ is true, so the definition demands $H(1) = 1 + 4 = 5$ and $H(1) = 3(1) = 3$ at the same time. One input with two outputs is not a function — the definition is broken before any arithmetic happens.

Step 2: repair with one character. Keep the top rule closed and open the bottom one: change $x \geq 1$ to $x > 1$. Now the conditions are $x \leq 1$ and $x > 1$: every real number satisfies exactly one.

Step 3: evaluate the repaired function at the boundary using the top rule, $x + 4$ at $x = 1$:

5

Step 4: the other legal repair — changing the top rule to $x < 1$ — also makes H a function, but then the bottom rule owns the boundary and $H(1) = 3$. Both repairs are valid definitions; they simply define different functions. The problem asked for the one where the top rule keeps the boundary.

5. (a) Why does K fail at $x = 0$, and what are two repairs? (b) Repair it so the top rule owns the boundary and give $K(0)$.

(a) *Open response — review by hand.*

Model answer. Both conditions are strict: the top rule needs $x < 0$ and the bottom rule needs $x > 0$. The input $x = 0$ satisfies neither, so no rule applies and $K(0)$ has no value at all — 0 is missing from the domain. This is the opposite fault to problem 4: an overlap gives the boundary two outputs, a gap gives it none.

Two repairs, each one character: change $x < 0$ to $x \leq 0$, so the top rule owns the boundary; or change $x > 0$ to $x \geq 0$, so the bottom rule does. Changing both would recreate the overlap of problem 4.

Look for: the observation that neither condition is satisfied at the boundary (not merely "the graph has a hole"), and two repairs that each close exactly one side.

(b) Take the first repair, $x \leq 0$, and evaluate the top rule $2x$ at $x = 0$:

$$0$$

6. A part passes when $|2x - 5| \leq 3$. Priya reports $1 < x < 4$ and says a part measuring exactly 1 fails. Find the endpoints and correct her.

Step 1: find the endpoints by solving the equation, because the absolute-value expression equals 3 exactly at the edges of the passing region.

Step 2: split the absolute value into its two cases: $2x - 5 = 3$ gives $2x = 8$, and $2x - 5 = -3$ gives $2x = 2$. So the endpoints are

$$x = 1 \text{ or } x = 4$$

Step 3: decide inclusion from the inequality sign, not from the picture. The condition is $|2x - 5| \leq 3$, and at $x = 1$ the left side is $|2(1) - 5| = 3$, which satisfies ≤ 3 . The same holds at $x = 4$.

Step 4: correct the report. Both endpoints pass, so the passing range is $1 \leq x \leq 4$, written with closed endpoints. Priya's $1 < x < 4$ would reject two parts that meet the specification exactly — the error costs real parts, which is why the sign matters.

7. Find k so that M has no jump at $x = 2$.

Step 1: write what "no jump" means at a boundary: the value the top rule approaches as x climbs to 2 must equal the value the bottom rule takes at $x = 2$. The bottom rule owns the boundary ($x \geq 2$), so its value is the one the function actually takes.

Step 2: compute the bottom rule's value there: $2^2 - 1 = 3$. Compute what the top rule approaches: $4(2) + k = 8 + k$.

Step 3: set them equal and solve: $8 + k = 3$, so

$$k = -5$$

Step 4: check. With $k = -5$ the top rule approaches $4(2) - 5 = 3$ and the bottom rule takes 3 ✓. Note what did *not* change: the bottom rule still owns $x = 2$. Matching the values removes the jump; it does not move the boundary.

8. Challenge. Where do $y = x - 1$ and $y = x^2 - 4x + 3$ meet, and which x needs the closed inequality?

Step 1: the graphs meet where both equations hold at once, so set the two expressions for y equal: $x - 1 = x^2 - 4x + 3$.

Step 2: collect on one side: $0 = x^2 - 5x + 4$, which factors as $(x - 1)(x - 4) = 0$.

Step 3: solve and find each y from the line: at $x = 1$, $y = 1 - 1 = 0$; at $x = 4$, $y = 4 - 1 = 3$.

The meeting points are

$(1, 0)$ and $(4, 3)$

Step 4: answer the boundary question. If the line is used first — that is, on the left-hand interval — then the switch happens at the larger meeting point, $x = 4$, and the line's condition must include it: $x \leq 4$, closed, with the parabola taking $x > 4$. Writing both as $x \leq 4$ would repeat problem 4's overlap; writing both strictly would repeat problem 5's gap. Because the curves meet there, either assignment gives the same output at $x = 4$ — but only one of them is a valid definition.